PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Barriers and facilitators of lung cancer screening uptake: Protocol of a mixed methods systematic review
AUTHORS	Lin, Yu-an; Hong, Yu ting; Chen, Bo ni; Xiao, Huimin; Huang, Fei fei

VERSION 1 – REVIEW

REVIEWER	Christos Chouaïd Centre Hospitalier Intercommunal de Creteil
REVIEW RETURNED	05-Aug-2021

GENERAL COMMENTS	Only a minor coment, define 3LCS" in abstract section

REVIEWER	Anna Snavely Wake Forest School of Medicine, Biostatistics and Data Science
REVIEW RETURNED	29-Aug-2021

GENERAL COMMENTS	Thank you for the opportunity to review this protocol for your mixed methods systematic review. While this systematic review has the potential to fill in some knowledge gaps surrounding barriers and facilitators for LCS, I have some concerns. 1. You are including quantitative assessments as part of your inclusion criteria. However, your analysis plans include only reporting frequencies. Why are you not including standard meta analysis methods to combine quantitative information across studies? And why are you limiting the data to statistically significant results? Non-significant results can be important too. 2. Risk is one of your subgroups of interest. How are you defining risk? Will the definition differ depending on the study? 3. Geographic location may be very important, particularly for heath care workers since health care systems can vary greatly in different countries. How will this be addressed in your work? 4. It would be helpful to have someone review the paper who can help with English language issues.

REVIEWER	HQ Nguyen Kaiser Permanente Southern California
REVIEW RETURNED	28-Oct-2021

GENERAL COMMENTS	This systematic review protocol paper attempts to use a socio-ecological framework to summarize multi-level barriers and facilitators to uptake of lung cancer screening

	The premise for conducting this review is that the 3 systematic reviews were either too narrow or only included studies from the US. Do the authors believe there is sufficient non-US, especially Chinese studies that were excluded from prior reviews to warrant this systematic review? While the review protocol is ambitious with using a socio-ecological framework to capture the multi-level barriers, this is ultimately dependent on the quality and quantity of the published papers. How do they propose to reconcile the many potentially disparate studies to fully synthesize the various barriers and facilitators since much of this is so "context" dependent? Do they anticipate barriers that transcend culture, geography, and how health care is organized and financed? Agree with authors' plans to stratify findings by screening time and risk status Review will appropriately include both qualitative and quantitative studies Could the authors describe their experience with performing such systematic reviews Page 12, line 37, please clarify this statement: "there is still a desperate lack of pragmatic clinical trials on LCS" What kind of pragmatic trial are the authors referring to—to prove effectiveness of LCS or to evaluate different implementation strategies to increase uptake of LCS in eligible patients? Recommend English language editing

REVIEWER	Theodore Holford Yale School of Public Health, Biostatistics
REVIEW RETURNED	10-Nov-2021

GENERAL COMMENTS	Low dose computed tomography is well recognized as a promising tool in the effort to reduce the devastating effects of lung cancer throughout the world. However, it has been a challenge to have this technology widely used, so the public health benefits could be greatly enhanced. This research proposal is an effort to better understand barriers and facilitators for lung cancer screening by bringing together and synthesizing results from the literature. The approach that will be used in this research will identify articles written in English and Chinese that have relevant information on barriers and facilitators for lung cancer screening. For this search, the authors will use methods that have similarly been used for meta-analysis studies, which seem to be appropriate for achieving completeness and reducing selection bias. This part of the proposal is well described and appropriate. Details on data to be collected on barriers and facilitators is less clear. It seems likely that these will be identified in the discussion section a paper, hence, the inclusion of a particular item would depend on whether the authors thought about it at the time of writing. In addition, many of these items would be described as part of the narrative and they might be challenging to classify for the analysis. It seems that this could be an important aspect of this research and the process that will be used for classification should

	be more clearly described. Missing from this discussion are plans for the statistical analysis of the data that will be collected. On p.10, l.37, it is stated that “a narrative analysis” will be used. It is not clear what this means. In general, the wide collection of data from existing studies is used to refine the overall estimate of an effect. However, in this case it is not clear what the effect will be because of the limited detail on what will be coded related to the barriers and facilitators.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1	
Only a minor coment, define 3LCS" in abstract section	Thanks for the reviewer’s comments. We have added the full names of LCS.
Reviewer: 2	
1. You are including quantitative assessments as part of your inclusion criteria. However, your analysis plans include only reporting frequencies. Why are you not including standard meta analysis methods to combine quantitative information across studies? And why are you limiting the data to statistically significant results? Non-significant results can be important too.	Thanks for the reviewer’s comments. We have detailed the data analysis and synthesis methods.
2. Risk is one of your subgroups of interest. How are you defining risk? Will the definition differ depending on the study?	Thanks for the reviewer’s comments. We have added the definition of risk in the section of data synthesis.
3. Geographic location may be very important, particularly for heath care workers since health care systems can vary greatly in different countries. How will this be addressed in your work?	We agreed with reviewers’ comments. Indeed, the barriers and facilitators of uptake LCS behavior might be transcend culture, geography, and how health care is organized and financed, which we have added in the section of discussion and will be explored in our review.
4. It would be helpful to have someone review the paper who can help with English language issues.	Thanks for reviewer’s comments. As suggested, the grammar has been checked by an experienced English editor.
Reviewer: 3	
The premise for conducting this review is that the 3 systematic reviews were either too narrow or only included studies from the US. Do the authors believe there is sufficient non-US, especially Chinese studies that were excluded from prior reviews to warrant this systematic review?	Thanks for reviewer’s comments. After we search literatures, we found that there is a certain number of non-US studies, especially Chinese studies, which may also provide importance evidence of barriers and facilitators of LCS practice.
While the review protocol is ambitious with using a socio-ecological framework to capture the multi-level barriers, this is ultimately dependent on the quality and	We agreed with reviewers’ comments. Indeed, the barriers and facilitators of uptake LCS behavior might be transcend

quantity of the published papers. How do they propose to reconcile the many potentially disparate studies to fully synthesize the various barriers and facilitators since much of this is so "context" dependent? Do they anticipate barriers that transcend culture, geography, and how health care is organized and financed?	culture, geography, and how health care is organized and financed, which we have added in the section of discussion.
Could the authors describe their experience with performing such systematic reviews	Thanks for reviewer's comments. We have described the experience of performing systematic review of authors in the manuscript.
Page 12, line 37, please clarify this statement: "there is still a desperate lack of pragmatic clinical trials on LCS" What kind of pragmatic trial are the authors referring to—to prove effectiveness of LCS or to evaluate different implementation strategies to increase uptake of LCS in eligible patients?	Thanks for reviewer's comments. We have described the pragmatic trial in the section of ethics and dissemination.
Recommend English language editing	Thanks for reviewer's comments. As suggested, the grammar has been checked by an experienced English editor.
Reviewer: 4	
Details on data to be collected on barriers and facilitators is less clear. It seems likely that these will be identified in the discussion section a paper, hence, the inclusion of a particular item would depend on whether the authors thought about it at the time of writing. In addition, many of these items would be described as part of the narrative and they might be challenging to classify for the analysis. It seems that this could be an important aspect of this research and the process that will be used for classification should be more clearly described.	Thanks for reviewer's comments. We have detailed the data to be collected on barriers and facilitators and how to data synthesis.
Missing from this discussion are plans for the statistical analysis of the data that will be collected. On p.10, l.37, it is stated that "a narrative analysis" will be used. It is not clear what this means. In general, the wide collection of data from existing studies is used to refine the overall estimate of an effect. However, in this case it is not clear what the effect will be because of the limited detail on what will be coded related to the barriers and facilitators.	Thanks for the reviewer's comments. We have detailed the data analysis and synthesis methods.

VERSION 2 – REVIEW

REVIEWER	Anna Snavely Wake Forest School of Medicine, Biostatistics and Data Science
REVIEW RETURNED	13-Dec-2021

GENERAL COMMENTS	The authors have addressed my previous comments.

REVIEWER	HQ Nguyen Kaiser Permanente Southern California
REVIEW RETURNED	01-Dec-2021

GENERAL COMMENTS	Although the responses to my earlier comments were addressed, I did not feel the authors were thorough with their revision especially regarding the question about how they intend to account for differences in barriers/facilitators across cultures, how health care organized, and local context.

REVIEWER	Theodore Holford Yale School of Public Health, Biostatistics
REVIEW RETURNED	16-Dec-2021

GENERAL COMMENTS	The protocol described in this manuscript is for a systematic review of literature related to lung cancer screening and the challenge in recruiting subjects who are eligible for low-dose CT. Study design calls for a search of English and Chinese databases up to the end of 2020. Hence, data collection appears to have already been completed, although the manuscript is not clear on the status of the work. Past tense is used when describing the work, which also suggests that the study is essentially complete, although no results are presented. If indeed the study is now complete, then the point of this manuscript describing the protocol would be moot. In their response to the previous review, the authors indicate that they have provided further detail on the statistical methods that will be used to summarize the data gleaned from the selected papers. However, the changes are modest and the approach to be used is still vague. For example, they propose to use “the social-ecological model” without explaining what that model is. Similarly, heterogeneity will be assessed using the “forest plot” without indicating where this plot is coming from. The I2 statistic will be used, but there is no indication of what this statistic will show. This aspect of the paper is still weak. A study with this design should not take a great deal of time to complete once the studies have been extracted from databases, as they have here. The data are already available to the authors, so all that is left is to complete the final manuscript. Therefore, I did not find this manuscript to be very helpful.

VERSION 2 – AUTHOR RESPONSE

Reviewer: 2	**Responses**
The authors have addressed my previous comments	Thanks for the reviewer’s comments.
Reviewer: 3	**Responses**
Although the responses to my earlier comments were addressed, I did not feel the authors were thorough with their revision especially regarding the question about how they intend to account for differences in barriers/facilitators across cultures, how health care	Thanks for the reviewer’s comments. We will search a large number of relevant information to understand the organizational form, culture and environment of health care in the country or region where the eligible studies were

organized, and local context.	conducted. Besides, we will further explore the barriers and facilitators in different cultural backgrounds, classify and summarize them so as to analyze their similarities and differences, and explore the underlying reasons, which have been described specifically in the article.
Reviewer: 4	**Responses**
1. The protocol described in this manuscript is for a systematic review of literature related to lung cancer screening and the challenge in recruiting subjects who are eligible for low-dose CT. Study design calls for a search of English and Chinese databases up to the end of 2020. Hence, data collection appears to have already been completed, although the manuscript is not clear on the status of the work. Past tense is used when describing the work, which also suggests that the study is essentially complete, although no results are presented. If indeed the study is now complete, then the point of this manuscript describing the protocol would be moot.	Thanks for the reviewer's comments. Our study did not start until the submission, and now is in the stage of data processing. We extended the retrieval time to May 31, 2021 for a second retrieval. The results showed that only one study was eligible. We will consider including this study in the discussion section. Besides, the past tense used in the article has been modified.
2. In their response to the previous review, the authors indicate that they have provided further detail on the statistical methods that will be used to summarize the data gleaned from the selected papers. However, the changes are modest and the approach to be used is still vague. For example, they propose to use"the social-ecological model"without explaining what that model is. Similarly, heterogeneity will be assessed using the "forest plot"without indicating where this plot is coming from. The I2 statistic will be used, but there is no indication of what this statistic will show. This aspect of the paper is still weak.	Thanks for the reviewer's comments. The barriers and facilitators for the uptake of LCS were synthesized according to the social-ecological model. That is, the factors were assigned to one of the four general categories: individual, interpersonal, health system, and socio/environmental. If the included articles can be analyzed quantitatively, statistical heterogeneity between studies will be assessed using the forest plot, chi-squared test and I^2 statistic, which have been described specifically in the article.
3. A study with this design should not take a great deal of time to complete once the studies have been extracted from databases, as they have here. The data are already available to the authors, so all that is left is to complete the final manuscript. Therefore, I did not find this manuscript to be very helpful.	Thanks for the reviewer's comments. To the best of our knowledge, this study will be the first systematic review to investigate the barriers and facilitators of lung cancer screening uptake at the level of both individuals and healthcare providers based on an evidence-based perspective. Additionally, our study will provide evidence of factors influencing lung cancer screening and confirm the social-ecological model, which will be conducive to further exploring the development of lung cancer screening programmes.